

Probability Density Functions

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First, an example

Before the abstract definition, fix the standard normal PDF $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ in mind. At $x = 0$ its height is $\phi(0) = 1/\sqrt{2\pi} \approx 0.399$. That value is greater than nothing in particular — but observe: a PDF *can* exceed 1 at a point. What must equal 1 is the *area* under the curve, not the heights:

$$\int_{-\infty}^{\infty} \phi(x) dx = 1.$$

Read aloud: the integral from minus infinity to infinity of phi of x, d x, equals one. Density measures probability per unit length; the bookkeeping rule is on the integral, not on pointwise values.

1 Definitions

Definition 1 (Probability density function). *Let X be a real-valued random variable on a probability space $(\Omega, \mathcal{F}, P)$. A non-negative, Lebesgue-integrable function $f_X : \mathbb{R} \rightarrow [0, \infty)$ is called the probability density function (PDF) of X if for every Borel set $B \subseteq \mathbb{R}$,*

$$P(X \in B) = \int_B f_X(x) dx.$$

Read aloud: the probability that X lies in B equals the integral over B of f sub X of x , d x . Equivalently, the law of X is absolutely continuous with respect to Lebesgue measure, and $f_X = dP_X/d\lambda$ is the Radon–Nikodym derivative.

Definition 2 (CDF–PDF relation). *The cumulative distribution function $F_X(x) = P(X \leq x)$ satisfies*

$$F_X(x) = \int_{-\infty}^x f_X(t) dt, \quad f_X(x) = F_X'(x) \text{ a.e.}$$

Read aloud: F sub X of x equals the integral from minus infinity to x of f sub X of t , d t ; and f sub X of x equals F sub X prime of x almost everywhere.

Proposition 1 (Normalization). *Any PDF f_X satisfies $\int_{-\infty}^{\infty} f_X(x) dx = 1$.*

Proof. $\int_{\mathbb{R}} f_X = P(X \in \mathbb{R}) = 1$. □

2 The Gaussian PDF

Define the *standard Gaussian density*

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right).$$

Read aloud: phi of x equals one over square-root of two pi, times e to the minus x squared over two. Clearly $\phi > 0$, so to show ϕ is a PDF we need only verify normalization.

Theorem 1 (Gaussian normalization). *The Gaussian PDF ϕ integrates to 1 over $\mathbb{R}$:*

$$\int_{-\infty}^{\infty} \phi(x) dx = 1.$$

Read aloud: the integral from minus infinity to infinity of phi of x, d x, equals one.

Proof. Let $I := \int_{-\infty}^{\infty} e^{-x^2/2} dx$. The integral is finite by comparison with a step function (or by direct integrability check). We compute I^2 via Fubini's theorem on the product:

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2/2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2/2} dy \right) = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2} dx dy.$$

Read aloud: I squared equals the product of two one-dimensional Gaussian integrals, which equals the double integral over $\mathbb{R}^2$ of e to the minus x squared plus y squared over two, $dx dy$. Switch to polar coordinates $x = r \cos \theta$, $y = r \sin \theta$, with Jacobian r :

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2/2} r dr d\theta.$$

Read aloud: I squared equals the integral from zero to two pi, of the integral from zero to infinity of e to the minus r squared over two, times r , $dr d\theta$. The inner integral evaluates by substitution $u = r^2/2$, $du = r dr$:

$$\int_0^{\infty} r e^{-r^2/2} dr = \int_0^{\infty} e^{-u} du = 1.$$

Read aloud: the integral from zero to infinity of $r e$ to the minus r squared over two, dr , equals the integral from zero to infinity of e to the minus u , du , which equals one. Therefore

$$I^2 = \int_0^{2\pi} 1 d\theta = 2\pi,$$

Read aloud: I squared equals the integral from zero to two pi of one, $d\theta$, which equals two pi. so $I = \sqrt{2\pi}$ (taking the positive root, as $e^{-x^2/2} > 0$). Hence

$$\int_{-\infty}^{\infty} \phi(x) dx = \frac{1}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = 1. \quad \square$$

Read aloud: the integral of phi of x , dx , equals one over square-root two pi, times square-root two pi, which equals one.

Proposition 2 (Gaussian CDF). *The standard Gaussian CDF $\Phi(x) = \int_{-\infty}^x \phi(t) dt$ admits no elementary closed form, but*

$$\Phi(x) = \frac{1}{2} \left(1 + \operatorname{erf}(x/\sqrt{2}) \right),$$

Read aloud: capital Phi of x equals one half times the quantity one plus the error function of x over square-root two. where $\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt$.

3 Remark on densities versus probabilities

A PDF can exceed 1. For example, the uniform PDF on $[0, 1/100]$ is $f(x) = 100$ on its support, yet $\int f = 1$. Density is probability *per unit length*, not probability itself; pointwise probability $P(X = x) = 0$ for any continuous random variable.